

A RATING FORMULATION FOR ORDERED RESPONSE CATEGORIES

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A rating response mechanism for ordered categories, which is related to the traditional threshold formulation but distinctively different from it, is formulated. In addition to the subject and item parameters two other sets of parameters, which can be interpreted in terms of thresholds on a latent continuum and discriminations at the thresholds, are obtained. These parameters are identified with the category coefficients and the scoring function of the Rasch model for polychotomous responses in which the latent trait is assumed uni-dimensional. In the case where the threshold discriminations are equal, the scoring of successive categories by the familiar assignment of successive integers is justified. In the case where distances between thresholds are also equal, a simple pattern of category coefficients is shown to follow.

Key words: logistic latent trait, Rasch model, thresholds, ordered categories.

1. Introduction

An interesting and perhaps puzzling feature of the Rasch model for polychotomous responses in which the latent trait is presumed uni-dimensional is that the category coefficient and scoring function parameters of the model have not been interpreted fully with respect to any response mechanism. The main purpose of this paper is to specify a rating mechanism which leads to such an interpretation. Andersen [1977] has derived certain statistical properties of the model and in a sense therefore, this paper is complementary to Andersen's.

We commence our development by tracing briefly the circumstances which have created this unusual case of uninterpreted parameters. We do this in Section 2 within the context of dichotomous response models. In Section 3 we outline generalizations of these models to polychotomous cases. In Section 4, and from the first stage of the traditional formulation, we propose a response mechanism which leads to the model identifiable with the Rasch model. In Section 5 we consider some implications of the obtained model and we summarize these in Section 6.

2. Dichotomous Response Models

In this section we highlight certain features of dichotomous response models necessary to place the subsequent development into context.

In latent trait theory for dichotomous test items, two distinct types of formulations seem to have evolved. The first, exemplified by Lord [1952] and Birnbaum [1968], has its origins in specifying plausible subject and item parameters together with a plausible probability distribution for a correct response. The second, exemplified by Rasch [1961, Note 1] and Andersen [1977], has its origins in specifying certain conditions for the estimation of subject and item parameters and then deducing the form of the possible probability distribution.

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In the former, a subject ability parameter β and two item parameters, the difficulty δ and discrimination α , are postulated. These parameters, found in traditional test theory and pertaining to the ubiquitous location and dispersion properties of distributions, are clearly plausible. Then if X is a random variable which takes the value $x = 1$ for a correct response and 0 otherwise, the probability distribution proposed by Lord is

$$(1) \quad p\{X = 1 \mid \beta, \delta, \alpha\} = \Phi(\alpha(\beta - \delta))$$

where Φ is the cumulative normal distribution with mean zero and variance unity. The distribution suggested by Birnbaum is similar taking the form

$$(2) \quad p\{X = 1 \mid \beta, \delta, \alpha\} = \Psi(\alpha(\beta - \delta))$$

where Ψ is the logistic function $\Psi(y) = \exp(y)/(1 + \exp(y))$.

The basic requirement specified by Rasch and Andersen is that the models have sufficient statistics for any subject and item parameters. For dichotomous responses, the Rasch model can take the form (2) except that it has no unknown discrimination parameter, that is,

$$(3) \quad p\{X = 1 \mid \beta, \delta\} = \Psi(\beta - \delta).$$

From experience with the first type of formulation, the parameters β and δ are then readily identified as the subject ability and item difficulty respectively. In fact, it is customary to consider the Rasch model as a special case of the Birnbaum model in which the common discrimination parameter is routinely absorbed into the other parameters as an arbitrary choice of unit. However, we must stress that Rasch does not arrive at his model by specializing the Birnbaum model but that he derives it from his requirement of sufficient statistics.

Partly because the models look so similar and partly because of the constant interplay between data, models, and the study of numerical methods, the distinctions between the two approaches have been blurred. However, they have different advantages and disadvantages which manifest themselves in correspondingly different ways.

An advantage in the first approach is that if one is not satisfied with the parameterization of a model, one can suggest new or restructured parameters in a way directed primarily by the response mechanism. Examples of this procedure are the addition of a guessing parameter to the item [Birnbaum, 1968], a guessing parameter to the subject [Waller, Note 4] and a discrimination parameter to the subject [Lumsden, 1977]. A disadvantage of this approach is that it may not be possible to estimate the parameters so introduced.

The advantage of the second approach, in which one is confined to a certain class of models, is that one knows in advance certain properties of the derived distributions. However, a possible disadvantage is that one may obtain a model which cannot be readily connected to any response mechanism generating the data. And this is the situation which has been reached with the Rasch model for polychotomous responses in which the item and subject parameters are assumed unidimensional.

3. *Unidimensional Models for Polychotomous Responses*

In this section we consider both the traditional and Rasch polychotomous response models for unidimensional traits. We follow the pattern of the previous section and once again only highlight certain features which will direct us to the required response formulation. To facilitate its development we note the following four points in advance.

First, we will find it convenient to shift from achievement testing situations to attitude testing for our illustrations. Thus instead of the dichotomous response to an item being correct or incorrect it becomes say agree or disagree. Then β and δ are respectively interpreted as a subject's attitude and an item's inherent intensity or affective value.

Secondly, we assume immediately an inherent ordering of responses. For example, instead of the response being agree or disagree, the classification may be extended to agree, neutral and disagree, where the ordering is evident. Thirdly, and accordingly, we extend the values that the random variable X can take. In particular, it will take the value $x = 0$ for disagree, $x = 1$ for neutral, and $x = 2$ for agree. At this stage no meaning other than the naming of the ordered categories is attached to these numbers. We recognize that other generalizations of the dichotomous random variable place a 1 in the category of the response and 0's elsewhere. However, as will be seen later, certain advantages are gained from the approach we adopt.

Fourthly, we develop the model using only the three categories suggested above and then generalize the result to any number of categories.

The Polychotomous Threshold Formulation

Examples of extensions of models (1) and (2) to the polychotomous situation in psychometrics are found in Samejima [1969] and Kolakowski and Bock [Note 3]. It primarily involves specifying additional parameters which correspond to an item's affective value. With the logistic distribution, although the same principles hold with the normal, a form of the model is

$$(4) \quad \begin{aligned} p\{X = 0 \mid \beta, \alpha, \tau_1, \tau_2\} &= 1 - \Psi(\alpha(\beta - \tau_1)), \\ p\{X = 1 \mid \beta, \alpha, \tau_1, \tau_2\} &= \Psi(\alpha(\beta - \tau_1)) - \Psi(\alpha(\beta - \tau_2)), \\ p\{X = 2 \mid \beta, \alpha, \tau_1, \tau_2\} &= \Psi(\alpha(\beta - \tau_2)). \end{aligned}$$

A continuous latent random variable with probability density Ψ' is supposed to be induced by the reaction of a person to an item. The parameters τ_1 and τ_2 , known as thresholds, designate boundaries on this variable and the response is then supposedly determined by the interval in which the value of this random variable falls. This extension is displayed in Figure 1. There is no problem in interpreting the new parameters τ_1 and τ_2 , which replace δ , as distances along the same latent continuum as β . In addition, there is no problem in preserving the order of categories as only one of the possible responses can occur.

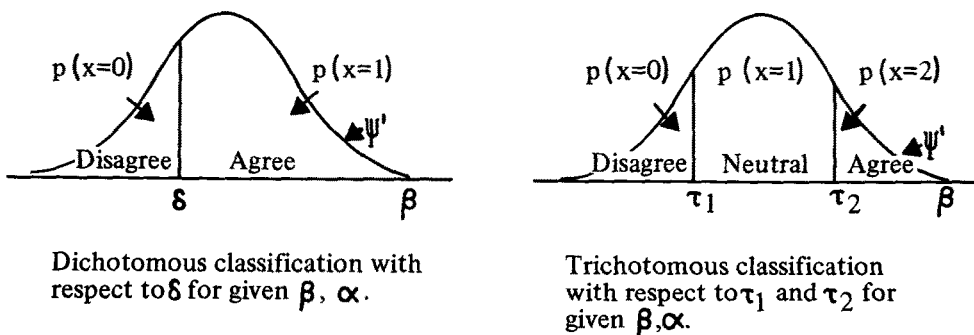


FIGURE 1

Latent processes and classification of the response for a threshold formulation.

The Polychotomous Rasch Formulation

In the first stage of the Rasch polychotomous formulation subjects and items are characterized by vectors. In the case of three categories, these may be characterized respectively by two independent parameters $\theta = (\theta_1, \theta_2)$ and $\sigma = (\sigma_1, \sigma_2)$. Then with the requirement of sufficient statistics, the deduced model may be expressed in the form

$$p\{X = 0 \mid \theta, \sigma\} = \frac{1}{\gamma(\theta, \sigma)},$$

$$(5) \quad p\{X = x \mid \theta, \sigma\} = \frac{\exp(\theta_x - \sigma_x)}{\gamma(\theta, \sigma)}, \quad x = 1, 2,$$

where

$$\gamma(\theta, \sigma) = 1 + \sum_{k=1}^2 \exp(\theta_k - \sigma_k),$$

[cf. Rasch, Note 1; Andersen, 1977]. This response model is clearly a straightforward generalization of (3).

In a specialization of (5) where θ and σ are considered functions of uni-dimensional parameters β and δ respectively, and still with the requirements of sufficient statistics for the subject and item parameters, the response model can be expressed as

$$p\{X = 0 \mid \beta, \delta, \kappa, \phi\} = \frac{1}{\gamma},$$

$$(6) \quad p\{X = x \mid \beta, \delta, \kappa, \phi\} = \frac{\exp[\kappa_x + \phi_x(\beta - \delta)]}{\gamma}, \quad x = 1, 2,$$

where

$$\gamma = \gamma(\beta, \delta, \kappa, \phi) = 1 + \sum_{k=1}^2 \exp[\kappa_k + \phi_k(\beta - \delta)].$$

Clearly, (6) is also a generalization of (3). The κ_k are the category coefficients and the ϕ_k the scoring functions. In the case of dichotomous responses, these specialize to $\kappa_1 = 0$ and $\phi_1 = 1$. (In a specialization of (5), which is more general than (6), the scoring function for the subject parameters is not the same as the scoring function for the item parameters. Because we wish to consider even greater specializations of (6), we commence with it as our most general case.)

Although κ and ϕ specialize readily in the dichotomous case, they are the terms which have not been interpreted in terms of a plausible response mechanism. Of course, ϕ is some kind of weighting factor which we would expect to increase as the order of the category increases and κ is some kind of location factor, but these observations do not lead us to the source of these parameters. Continuing with the approach of specifying requirements and deducing the form of models, Andersen [1977] lists certain ordering requirements of relevant transformations and then arrives at an "equi-distant" scoring function. Unfortunately, this still does not provide interpretations which can be useful in constructing items or response categories which subscribe to that particular scoring function. Furthermore, nothing at all seems to have been said regarding any characteristics that the category coefficients might have. In the next section we derive the distribution in which these parameters may be identified.

4. A Rating Formulation

The successful identification of the Rasch model parameters with those from other formulations in the dichotomous case suggests that there may also be connections among them in the polychotomous case. Accordingly, we draw extensively on the threshold formulation. However, we will also make modifications required by the Rasch specifications, and to distinguish it from the threshold formulation, we will call it a *rating formulation*.

Suppose then that when a subject encounters an item, a continuous random variable is induced as in the threshold formulation but that its value is governed by both β and δ . Suppose also that threshold values τ_1 and τ_2 exist as before where $\tau_2 > \tau_1$. We then consider that δ and τ enter the probability distribution we specify in the form $(\delta + \tau)$. The specification is analogous to the threshold formulation shown in Figure 1 except that the item retains an identifiable parameter. This is the first difference between the two formulations and the idea is that the threshold value *qualifies* the affective value of the item, or the other way around. This qualification may show for example that an agree response to an item with a moderate affective value is equivalent to a neutral response to an item of high affective value. In achievement testing, it corresponds to a possible equality between an excellent performance on a moderately difficult task and a moderate performance on a very difficult one. It also implies that each threshold retains the same value with respect to all items, and again, the other way around.

It is evident that the specification reflects a certain symmetry between the item and threshold parameters. This symmetry has interesting consequences which are explored in more detail in Section 5.

Now because we have specified that each value τ qualifies δ , we will consider a separate response process with respect to each threshold for a given item. This is the second distinction between the threshold and rating formulations since in the former, only one process is considered. With respect to the two thresholds τ_1 and τ_2 we introduce for the moment two independent Bernoulli random variables, X_1 which takes the value $x_1 = 1$ when the first threshold is passed and 0 otherwise, and X_2 which takes the value $x_2 = 1$ when the second threshold is passed, and 0 otherwise. Using this notation and invoking the logistic distribution function, which appears in the Rasch model and which we portrayed in Figure 1 with respect to the threshold formulation, we may write the explicit probabilities

$$(7) \quad \begin{aligned} p\{X_1 = 0 \mid \beta, \delta, \alpha, \tau_1\} &= \frac{1}{\eta_1}, \\ p\{X_1 = 1 \mid \beta, \delta, \alpha, \tau_1\} &= \frac{\exp [\alpha\{\beta - (\delta + \tau_1)\}]}{\eta_1}, \end{aligned}$$

with respect to τ_1 where $\eta_1 = \eta_1(\beta, \delta, \alpha, \tau_1) = 1 + \exp [\alpha\{\beta - (\delta + \tau_1)\}]$. We provide this explicit notation because we will find it convenient in later generalizations.

The symmetry between δ and τ in the above expressions suggests that the discrimination α which pertains to a specific item in models (1) and (2) may also vary from threshold to threshold. For example, in response to an item which involves thresholds between neutral, agree, and strongly agree categories say, it may be easier to discriminate between neutral and agree than between agree and strongly agree. Or in rating a performance into one of the three categories fail, pass and credit pass, it may be easier to discriminate between fail and pass than between pass and credit pass. Therefore, qualifying the discrimination parameter by its subscript, (7) becomes

$$(8a) \quad \begin{aligned} p\{X_1 = 0 \mid \beta, \delta, \alpha_1, \tau_1\} &= \frac{1}{\eta_1}, \\ p\{X_1 = 1 \mid \beta, \delta, \alpha_1, \tau_1\} &= \frac{\exp [\alpha_1 \{\beta - (\delta + \tau_1)\}]}{\eta_1}. \end{aligned}$$

Similarly, with respect to the *separate* response process at τ_2

$$(8b) \quad \begin{aligned} p\{X_2 = 0 \mid \beta, \delta, \alpha_2, \tau_2\} &= \frac{1}{\eta_2}, \\ p\{X_2 = 1 \mid \beta, \delta, \alpha_2, \tau_2\} &= \frac{\exp [\alpha_2 \{\beta - (\delta + \tau_2)\}]}{\eta_2}. \end{aligned}$$

Figure 2 displays the possible response processes.

Although we have separated the response processes with respect to the two thresholds, if we require a single response we must bring them together somehow. It is at this point that our formulation has its third and most distinctive difference from the traditional threshold formulation. Specifically, it concerns the process by which the ordering of the response categories is invoked and preserved.

The separation of the processes with respect to the thresholds suggests that they may be experimentally independent. One can certainly envisage an encounter of a subject with an item with respect to the disagree versus neutral threshold on one occasion and the same person with the same item but with respect to the neutral versus agree threshold on a separate occasion. If these occasions were sufficiently spaced in time, we might find them statistically independent. In fact let us entertain the idea that they are independent at least *instantaneously* when the subject first encounters the item and its two qualifying thresholds. In that case, the entire sample space for the responses to the two thresholds commences as the set

$$\Omega = \{(0, 0), (1, 0), (1, 1), (0, 1)\},$$

where the first member of each ordered pair of the elements in Ω reflects the response to the first threshold, and the corresponding response probabilities are given by

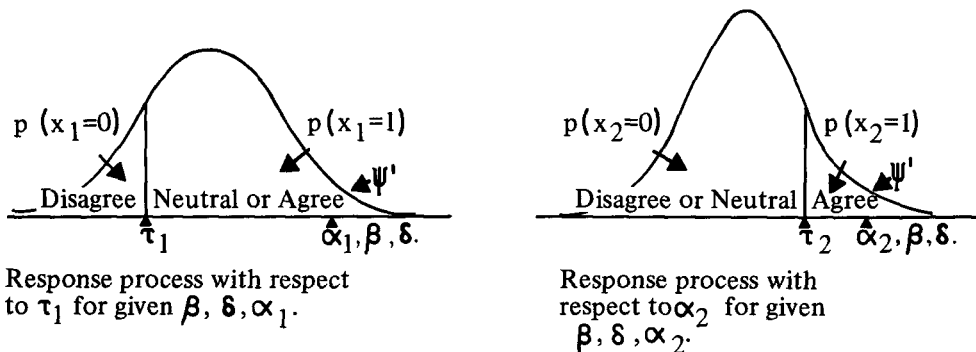


FIGURE 2
Latent processes and classification of the response for a rating formulation.

$$\begin{aligned}
 p\{(0, 0)\} &= \frac{1}{\eta_1 \eta_2}, \\
 p\{(1, 0)\} &= \frac{\exp [\alpha_1 \{\beta - (\delta + \tau_1)\}]}{\eta_1 \eta_2}, \\
 p\{(1, 1)\} &= \frac{\exp [\alpha_1 \{\beta - (\delta + \tau_1)\} + \alpha_2 \{\beta - (\delta + \tau_2)\}]}{\eta_1 \eta_2},
 \end{aligned}$$

and

$$p\{(0, 1)\} = \frac{\exp [\alpha_2 \{\beta - (\delta + \tau_2)\}]}{\eta_1 \eta_2}.$$

Now let us suppose that after the instantaneous reaction in the space Ω , that the subject brings the two processes together. To bring these together, we presume he recognizes that Ω is composed of the set $\Omega' = \{(0, 0), (1, 0), (1, 1)\}$ of legitimate responses and its complement the set $\{(0, 1)\}$ containing an illegitimate response. We presume that he recognizes that $\{(0, 1)\}$ is not legitimate because it would imply the inconsistency that his response reflects a position simultaneously below τ_1 and above τ_2 where $\tau_1 < \tau_2$. That is, he recognizes the ordering of the categories. Therefore we must suppose that even when the event in the complement of Ω' has happened instantaneously, that the response is reconsidered and redistributed in Ω' .

Our aim in the above considerations is not to discourse fully on the psychological mechanisms of rating. Instead it is intended to point out a possible conflicting situation for the rater. Nor are we in a position to consider the actual psychological mechanism behind any resolution of this possible conflict. However, if the subject makes a response, we may presume that he has resolved the conflict in some way. In that case we suggest the following effect of its resolution.

We make the plausible assumption that the original occurrences of illegitimate responses are redistributed in such a way that the legitimate ones retain the same relative probabilities as they had in the original space Ω . However, this is formally equivalent to considering a conditional distribution with respect to the legitimate subspace Ω' and this is obtained simply by normalizing the elements of Ω' to sum to unity.

After some simple algebra, we obtain

$$\begin{aligned}
 p\{(0, 0) \mid \beta, \delta, \alpha, \tau\} &= \frac{1}{\gamma}, \\
 (9) \quad p\{(1, 0) \mid \beta, \delta, \alpha, \tau\} &= \frac{1}{\gamma} \exp [\alpha_1 \{\beta - (\delta + \tau_1)\}], \\
 p\{(1, 1) \mid \beta, \delta, \alpha, \tau\} &= \frac{1}{\gamma} \exp [\alpha_1 \{\beta - (\delta + \tau_1)\} + \alpha_2 \{\beta - (\delta + \tau_2)\}],
 \end{aligned}$$

where

$$\begin{aligned}
 \gamma = \gamma(\beta, \delta, \alpha, \tau) &= 1 + \exp [\alpha_1 \{\beta - (\delta + \tau_1)\}] \\
 &\quad + \exp [\alpha_1 \{\beta - (\delta + \tau_1)\} + \alpha_2 \{\beta - (\delta + \tau_2)\}].
 \end{aligned}$$

Although (9) looks cumbersome, it can be simplified considerably. First, we reintroduce our original random variable X and observe that its values are connected to the events according to

$$\begin{aligned}
 x &= 0 \quad \text{for } (0, 0), \\
 x &= 1 \quad \text{for } (1, 0),
 \end{aligned}$$

and

$$x = 2 \quad \text{for} \quad (1, 1).$$

An advantage of our definition of X over that of a vector involving 0's and a 1 is now apparent from the above list. The extension to any number of thresholds is obvious, the value x indicating the number of ordered thresholds passed. Appropriately, $x = 0$ indicates that no threshold has been passed.

Secondly, we observe that

$$\alpha_1\{\beta - (\delta + \tau_1)\} = \{-\alpha_1\tau_1 + \alpha_1(\beta - \delta)\},$$

and that

$$\alpha_1\{\beta - (\delta + \tau_1)\} + \alpha_2\{\beta - (\delta + \tau_2)\} = \{-\alpha_1\tau_1 - \alpha_2\tau_2 + (\alpha_1 + \alpha_2)(\beta - \delta)\}.$$

Thirdly, we define the α and τ combinations according to the simple recursive relations

$$\begin{aligned} \phi_1 &= \alpha_1, \\ \kappa_1 &= -\alpha_1\tau_1, \\ (10) \quad \phi_2 &= \phi_1 + \alpha_2 = \alpha_1 + \alpha_2 \\ \kappa_2 &= \kappa_1 - \alpha_2\tau_2 = -\alpha_1\tau_1 - \alpha_2\tau_2. \end{aligned}$$

With these observations and definitions, (9) simplifies readily to

$$\begin{aligned} p\{X = 0 \mid \beta, \delta, \kappa, \phi\} &= \frac{1}{\gamma} \\ (11) \quad p\{X = 1 \mid \beta, \delta, \kappa, \phi\} &= \frac{1}{\gamma} \exp [\kappa_1 + \phi_1(\beta - \delta)] \\ p\{X = 2 \mid \beta, \delta, \kappa, \phi\} &= \frac{1}{\gamma} \exp [\kappa_2 + \phi_2(\beta - \delta)] \end{aligned}$$

where

$$\gamma = \gamma(\beta, \delta, \kappa, \phi) = 1 + \sum_{k=1}^g \exp [\kappa_k + \phi_k(\beta - \delta)].$$

However, not only have we simplified (9) considerably, but (11) is identical to (6) in which we wanted to interpret the parameters κ and ϕ . Thus through (10), we have identified these parameters in terms of the familiar threshold and discrimination concepts.

5. Implications of the Rating Formulation

We now consider some implications and specializations of this formulation.

Case 1: Only Threshold Discriminations Retained

In the development so far we have only considered the case of one subject responding to one item in one of a number of categories. In that case we need only to subscript the category discrimination parameter by the category index, e.g. α_k for the k^{th} threshold. However, if we consider that the subject responds to more than one item and suppose that items have different discriminations, then we might subscript α by i as well as k to α_{ik} . It would then be possible, using approaches found in the threshold formulation, to resolve α_{ik} into a part specific to the item and a part specific to the category. Since we are interpreting a Rasch model which requires that the items have the same discriminations, we will not pursue this idea. Therefore we may interpret the formulation to require that

each threshold discriminates in the same way irrespective of which item the threshold qualifies.

Without further elaboration, we generalize (11) to m thresholds to obtain

$$(12) \quad p\{X = x \mid \beta, \delta, \kappa, \phi\} = \frac{1}{\gamma} \exp [\kappa_x + \phi_x(\beta - \delta)],$$

where

$$\gamma = \gamma(\beta, \delta, \kappa, \phi) = \sum_{k=0}^m \exp [\kappa_k + \phi_k(\beta - \delta)],$$

$$\kappa_x = \phi_x = 0 \quad \text{for } x = 0,$$

and

$$\kappa_x = - \sum_{k=1}^x \alpha_k \tau_k \quad \text{and} \quad \phi_x = \sum_{k=1}^x \alpha_k, \quad x = 1, m.$$

Equation (12) is the general form of the rating response model.

Case 2: The Integral Scoring Function

With the retention of identical discriminations among items, a parallel consideration regarding the possibility of identical discriminations among threshold arises. For this first specialization of (12), let $\alpha_1 = \alpha_2 = \dots = \alpha_m = \alpha$. Then

$$\kappa_x = - \sum_{k=1}^x \alpha \tau_k \quad \text{and} \quad \phi_x = \alpha x.$$

Absorbing the constant discrimination α at the threshold into the subject, item and threshold parameters, as is done with the constant *item* discrimination in the dichotomous model, gives

$$(13) \quad p\{X = x \mid \beta, \delta, \kappa\} = \frac{1}{\gamma} \exp [\kappa_x + x(\beta - \delta)],$$

where

$$\gamma = \gamma(\beta, \delta, \kappa) = \sum_{k=0}^m \exp [\kappa_k + k(\beta - \delta)],$$

$$\kappa_0 = 0 \quad \text{and} \quad \kappa_x = - \sum_{k=1}^x \tau_k, \quad x = 1, m.$$

Since this specification holds for all x , $x = 0, \dots, m$, we see that, commencing with 0, the successive categories are scored with successive integers. Thus in (13), the form of our second case, we have an *integral scoring function*. In addition, the value of the random variable appears conveniently in the expression of the probability distribution and therefore the specification of the random variable X to denote responses in successive categories by successive integers becomes even more advantageous.

In this context, two further interesting points may be observed with respect to the integral scoring function. First, if a subject responds to n items and his score on each item i is x_i , then the sufficient statistic r of β [Andersen, 1977] is the total score $r = \sum_i x_i$. That is, the sufficient statistic is simply the number of thresholds, with respect to all items, that the subject has passed. In the case where there is only one threshold, which is then absorbed into the item parameters and the response is dichotomous, the sufficient statistic specializes to the total number of items which the subject has passed. Analogously, when there is

only one item, it specializes to the number of thresholds which the subject has passed. This analogy clearly reflects the symmetry we introduced between the items and thresholds.

The second and related point is that the integral scoring function does not arise from any references to distances. Instead, it arises from the number of equally discriminating thresholds passed. This is in contrast to certain traditional formulations where the so called "equidistant" scoring is considered justified only if the distances between successive categories are equal. In the model under discussion the symmetry between the item and threshold parameters is again manifested. Thus the simple summing of items passed in the dichotomous response case does not depend on any distance properties among item parameters δ , which are accounted for separately in the model as are the threshold parameters τ , but on the equality of item discriminations.

A further consequence of the symmetry between items and thresholds with respect to the sufficient statistics, which is established by Andersen [1977], can be seen readily. Briefly, if item discriminations α_i are assumed in the dichotomous case, then the "sufficient" statistic for a subject's attitude is given by $r = \sum_i \alpha_i x_i$ [Birnbau, 1968]. If the α_i are different, then in general the range space R , $r \in R$ is in one-to-one correspondence with the observation space χ , $x_i \in \chi$. Therefore when the parameter β of a specified subject is to be estimated independently of any population to which he may belong, this statistic provides no effective data reduction. The value associated with the response to each item must be known even if all discriminations are known in advance. Analogously, in the case where the threshold discriminations are free to be different, the sufficient statistic for a subject's attitude is given by

$$r = \sum_x \phi_x t_x$$

where t_x is the number of times the subject has responded in category x with respect to all items. If $\phi_x - \phi_{x-1} \neq \phi_{x+1} - \phi_x$ for all x , $x = 1, m$, then Andersen shows that in general the range space R , $r \in R$ has as many elements as the space T , $t \in T$ and therefore that there is again no effective data reduction beyond the observed pattern t_x . However, if the threshold discriminations are equal then $\phi_x - \phi_{x-1} = \phi_{x+1} - \phi_x = 1$ and as mentioned earlier the sufficient statistic for β becomes $r = \sum_i x_i$. In this case there is an effective data reduction.

Another observation of Andersen's, which concerns the case where two successive scoring functions are equal, becomes readily interpretable. Andersen provides an illustration and concludes that the two corresponding successive categories may then be pooled because the choice of category between these two is random. To see that this conclusion is consistent with our development, consider again the general recursive relationship, exemplified in (10), which relates the threshold discriminations and the scoring functions. We have $\phi_{x+1} = \phi_x + \alpha_{x+1}$. Now suppose that two successive scoring functions are equal, say $\phi_{x+1} = \phi_x$ for some x . This will occur when $\alpha_{x+1} = 0$. However, this simply implies that the $(x+1)^{\text{th}}$ threshold does not discriminate. Accordingly, the categories on either side of it may be pooled.

In connection with equal discriminations at thresholds, we note that the use of the term 'equidistant' for the scoring function where $\phi_x - \phi_{x-1} = \phi_{x+1} - \phi_x$ is a little misleading. This function is not related to distances but to discriminations. In the next section we do consider the case where actual distances between thresholds are equal.

Finally, in Figure 3 the category characteristic curves for this important case are shown for three symmetrically placed thresholds as a function of the combined subject and item parameter $\mu = (\beta - \delta)$. Three related, consistent and intuitively appealing properties, which can be shown readily algebraically, are revealed in Figure 3. Firstly, the greater the difference between the subject and item parameter, the greater the probability of a high score. Secondly, the score with the highest probability corresponds to the interval between

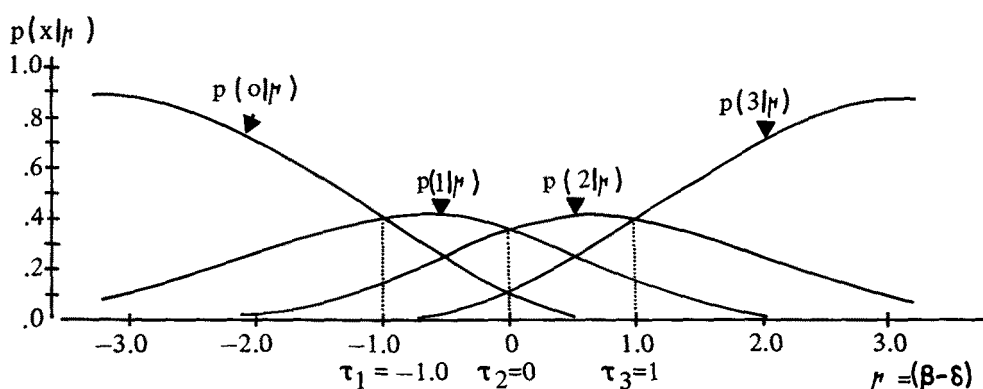


FIGURE 3

Category characteristic curves for the integral scoring function and three thresholds as a function of the difference between subject and item location.

thresholds in which the difference $\mu = (\beta - \delta)$ falls. Thirdly, and consequently, when μ is equal to a threshold value, the probabilities of the responses in the categories on either side of the threshold are equal.

Case 3: Equidistant Thresholds

Our interpretations so far have focussed on a specialization of the threshold discriminations. We retain that specialization but now consider the case where the distances between successive thresholds are all unity. In order to simplify the subsequent exposition, it will be expedient to digress a little and show that the threshold values can always be centered to have a mean of 0.

First, we recall that in the exponent $\kappa_x + x(\beta - \delta)$ in the expressions of (13), $\kappa_0 = 0$. Secondly, we expand each of the remaining terms according to

$$\kappa_x + x(\beta - \delta) = \{-\tau_1 + (\beta - \delta) + \cdots \{-\tau_x + (\beta - \delta)\}.$$

If

$$\sum_{k=1}^m \tau_k / m = \bar{\tau} \neq 0, \quad \text{let } \tau_k^* = \tau_k - \bar{\tau}, \quad \text{in which case } \sum_{k=1}^m \tau_k^* = 0.$$

Then defining β^* according to $\beta^* = \beta - \bar{\tau}$, gives

$$\kappa_x + x(\beta - \delta) = \{-\tau_1^* + (\beta^* - \delta) + \cdots \{-\tau_x^* + (\beta^* - \delta)\}.$$

Thus a constant can be absorbed into β so that

$$\sum_{k=1}^m \tau_k^* = 0.$$

We now return to (13), but first we drop the “*” from the new parameters and assume for convenience that the τ have already been centered to a mean of zero. Then setting the distances between thresholds equal to unity making $\tau_k - \tau_{k-1} = 1$ for $k > 1$, provides the

TABLE 1

Category Coefficients for Equi-distant Thresholds for
 $m = 1, 6$.

m	Category Coefficients	Corresponding Values
1		
2	κ_0 κ_1	$\begin{bmatrix} 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}^*$
3	κ_0 κ_1 κ_2	$\begin{bmatrix} 0 & 2 & 2 & 0 \\ 0 & 3 & 4 & 3 & 0 \end{bmatrix}^*$
4	κ_0 κ_1 κ_2 κ_3	$\begin{bmatrix} 0 & 4 & 6 & 6 & 4 & 0 \\ 0 & 5 & 8 & 9 & 8 & 5 & 0 \end{bmatrix}^*$
5	κ_0 κ_1 κ_2 κ_3 κ_4	
6	κ_0 κ_1 κ_2 κ_3 κ_4 κ_5 κ_6	

*Each value in the table is divided by 2 as indicated.

following pattern of threshold values for increasing values of m :

$$\begin{array}{l}
 m = 1 \\
 m = 2 \\
 m = 3 \\
 m = 4 \\
 m = 5
 \end{array}
 \begin{bmatrix}
 & & \tau_1 & & \\
 & & \tau_1 & \tau_2 & \\
 & \tau_1 & \tau_2 & \tau_3 & \\
 \tau_1 & \tau_2 & \tau_3 & \tau_4 & \\
 \tau_1 & \tau_2 & \tau_3 & \tau_4 & \tau_5
 \end{bmatrix}
 \rightarrow \frac{1}{2}
 \begin{bmatrix}
 & & 0 & & \\
 & & -1 & 1 & \\
 & -2 & 0 & 2 & \\
 -3 & -1 & 1 & 3 & \\
 -4 & -2 & 0 & 2 & 4
 \end{bmatrix}$$

However, recognizing that $\kappa_0 = 0$ and that

$$\kappa_x = -\sum_{k=1}^x \tau_k \quad \text{for } x \neq \emptyset,$$

enables us to extract a corresponding pattern of values for the category coefficients κ_x , $x = 0, m$. These, analogous to Pascal's triangle for binomial coefficients, are shown in Table 1 for $m = 1, 6$. It can in fact be readily shown that $\kappa_x = x(m-x)/2$.

Therefore for this second specialization,

$$(14) \quad p\{X = x \mid \beta, \delta, m\} = \frac{1}{\gamma} \exp \left[\frac{x(m-x)}{2} + x(\beta - \delta) \right]$$

where

$$\gamma = \gamma(\beta, \delta, m) = \sum_{k=0}^m \exp \left[\frac{k(m-k)}{2} + k(\beta - \delta) \right].$$

6. Summary and Remarks

A rating formulation, different from the traditional threshold formulation but set in a similar context, is proposed. The main difference between the rating and threshold formulations is in the proposed mechanism which preserves the ordering of the response categories.

The resultant response model is identified with the Rasch model for polychotomous responses in which the latent trait is assumed uni-dimensional. The category coefficients and the scoring functions of the Rasch model are interpreted in terms of thresholds on the

latent continuum and in terms of discriminations at the thresholds. However, if the threshold discriminations are not equal, then the complete pattern of category responses can be recovered from the statistic used to estimate a subject's attitude. Accordingly, no data reduction beyond the response pattern is effected. On the other hand, the model may be useful for diagnostic analyses of items and categories.

By specializing the threshold discriminations to be equal, it is shown that the scoring function simplifies to the scoring of successive categories with successive integers. As a result, the total score of a subject with respect to all items becomes the sufficient statistic for the attitude estimate and it does provide an effective data reduction. In a further specialization, in which unit distances between successive thresholds are assumed, it is demonstrated that the category coefficients also take on a very simple functional form.

Statistical procedures for parameter estimation and tests of fit for the various models are not discussed. However, without an explicit interpretation of the category coefficients and the scoring functions, these procedures have already received attention in the literature. For example, Andersen [1973] describes and illustrates the estimation of the scoring function while Andrich [Note 2] shows an application of the model in which the category coefficients are estimated when the integral scoring function is assumed. Adaptations of these procedures to the above interpretations of the parameters should not be difficult, but a discussion of that topic is considered outside the scope of this paper.

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